

Ch1

给出 PDE 的基本概念, 符号, 对方程的分类
研究 PDE 的主要目标.

§1.1 偏微分方程的基本概念

方程: 未知数组成的等式

常微分方程: 一元未知函数及其导数组成的方程 (ODE)

未知的多元函数及其偏导数构成的方程, 称偏微分方程 (PDE.)

PDE 方程组: 至少两个以上的偏微分方程组成的系统.

符号: 区域 $\Omega \subseteq \mathbb{R}^n$, $n \geq 2$, $x \in \Omega$, $x = (x_1, x_2, \dots, x_n)$ 设 $u(x): \Omega \rightarrow \mathbb{R}$ 为一函数.

$$\frac{\partial u}{\partial x_i} \quad i=1,2,\dots,n, \quad \frac{\partial^2 u}{\partial x_i \partial x_j}, \quad \frac{\partial^2 u}{\partial x_i^2} \quad i,j=1,2,\dots,n$$

通常简记为

$$\partial_i u = u_{x_i} = \frac{\partial u}{\partial x_i} \quad \partial_{ij} u = u_{x_i x_j} = u_{ji} = \frac{\partial^2 u}{\partial x_i \partial x_j}$$

一般地, 对 $k \in \mathbb{N}$,

$$\frac{\partial^k u}{\partial x_{i_1} \partial x_{i_2} \dots \partial x_{i_k}}, \quad i_1, i_2, \dots, i_k \in \{1, 2, \dots, n\} \quad (\text{有可能 } i_l = i_m, l \neq m)$$

引入多重指标

$$\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n) \text{ 多重指标, } \alpha_i \in \mathbb{N}, \quad |\alpha| = \alpha_1 + \alpha_2 + \dots + \alpha_n.$$

$$\frac{\partial^{|\alpha|} u}{\partial x_1^{\alpha_1} \partial x_2^{\alpha_2} \dots \partial x_n^{\alpha_n}}$$

梯度 gradient Du , ∇u

$$Du = \left(\frac{\partial u}{\partial x_1}, \frac{\partial u}{\partial x_2}, \dots, \frac{\partial u}{\partial x_n} \right) = (u_{x_1}, u_{x_2}, \dots, u_{x_n})$$

* Hessian 矩阵 D^2u , $\nabla^2 u$

$$D^2u = \begin{pmatrix} \frac{\partial^2 u}{\partial x_1^2} & \frac{\partial^2 u}{\partial x_1 \partial x_2} & \dots & \frac{\partial^2 u}{\partial x_1 \partial x_n} \\ \frac{\partial^2 u}{\partial x_2 \partial x_1} & \frac{\partial^2 u}{\partial x_2^2} & \dots & \frac{\partial^2 u}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 u}{\partial x_n \partial x_1} & \frac{\partial^2 u}{\partial x_n \partial x_2} & \dots & \frac{\partial^2 u}{\partial x_n^2} \end{pmatrix}$$

$$|D^2u| = \left(\sum_{i,j=1}^n \left| \frac{\partial^2 u}{\partial x_i \partial x_j} \right|^2 \right)^{\frac{1}{2}}$$

* 散度 divergence 设 $\vec{F}(x) = (F_1(x), \dots, F_n(x)): \Omega \rightarrow \mathbb{R}^n$ 为向量场, 则

* 散度 divergence 设 $\vec{F}(x) = (F_1(x), \dots, F_n(x)) : \Omega \rightarrow \mathbb{R}^n$ 为向量值函数.

$$\operatorname{div} \vec{F} = \frac{\partial F_1}{\partial x_1} + \dots + \frac{\partial F_n}{\partial x_n} = \sum_{i=1}^n \frac{\partial F_i}{\partial x_i} \quad (\nabla \cdot \vec{F})$$

$$\Delta u = \sum_{i=1}^n \frac{\partial^2 u}{\partial x_i^2} = \operatorname{tr}(D^2 u) = \operatorname{div}(\nabla u) = \nabla \cdot (\nabla u)$$

* PDE 的阶数: PDE 中未知函数偏导数的最高次数.

* PDE 的解: 满足 PDE 的函数.

例. 对于 k 阶 PDE. $F(D^k u, D^{k-1} u, \dots, u) = 0$

方程的解 u 的光滑性, 有时有 k 阶偏导, 有时没有 k 阶导数.

解不唯一时, 有 k 阶可导的解, 也有达不到 k 阶可导的解.

(光滑解, 古典解, 解析解, 弱解, 粘性解)

* 定义空间 $C^k(\Omega)$: Ω 上所有具有 k 阶连续偏导数的函数组成的集合 (线性空间)

(由于求导运算是线性的, 所以是线性空间)

$$C(\Omega) = C^0(\Omega) \quad u \in C(\Omega) \quad \text{模} \quad \|u\|_{C(\Omega)} = \sup_{x \in \Omega} |u(x)|$$

$$u \in C^k(\Omega) \quad \text{模} \quad \|u\|_{C^k(\Omega)} = \sup_{x \in \Omega} |u(x)| + \sum_{|\alpha|=1}^k \sup_{x \in \Omega} |D^\alpha u(x)|$$

* $C_0^k(\Omega)$, $C_c^k(\Omega)$ 具有紧支集的 $C^k(\Omega)$.

$$\text{支集 (support)} \quad u \in C(\Omega) \quad \operatorname{supp} u := \overline{\{x \in \Omega : u(x) \neq 0\}}$$

$C_0^k(\Omega)$ 具有紧支集的 k 阶连续偏导数的函数类. (紧: 支集在 $\Omega \subseteq \mathbb{R}^n$ 中有界)

$$C^\infty(\Omega) := \bigcap_{k=1}^{\infty} C^k(\Omega) \quad (\text{不是线赋范空间, 但其中有收敛性})$$

$C_0^\infty(\Omega)$ 具有紧支集的任意阶偏导数连续的函数类.

* 古典解 (classical solution 经典解)

对于 PDE $F(D^k u, D^{k-1} u, \dots, u, x) = 0 \quad x \in \Omega$. 若解 $u \in C^k(\Omega)$ 则 u 是古典解.

线性 $\sum_{|\alpha| \leq k} a_\alpha(x) D^\alpha u = f(x) \quad \text{例: Poisson 方程: } -\Delta u = f(x) \quad f \text{ 为给定函数.}$

PDE 关于未知 u 的方程

PDE关于线性性的分类

线性 $\sum_{|\alpha| \leq k} a_\alpha(x) D^\alpha u = f(x)$ 例: Poisson 方程: $-\Delta u = f(x)$ f 为给定函数.

非线性

$$D(Au(x) + Bv(x))$$

$$= A Du + B Dv$$

对未知函数的运算是否是线性运算.

1. 半线性

$$\sum_{|\alpha| \leq k} a_\alpha(x) D^\alpha u = f(D^{k-1}u, \dots, Du, u, x)$$

例: Schrödinger 方程: $-\Delta u + V(x)u = f(u)$

2. 拟线性

$$\sum_{|\alpha| \leq k} a_\alpha(D^{k-1}u, \dots, Du, u, x) D^\alpha u = f(D^{k-1}u, \dots, Du, u, x)$$

例: $-\Delta u^2 - \Delta u + V(x)u = f(u)$

$$\text{其中 } \Delta u^2 = \operatorname{div}(\nabla u^2) = \operatorname{div}(2u \nabla u) = 2|\nabla u|^2 + 2u \Delta u$$

3. 完全非线性: 关于最高阶偏导数是非线性的.

例: Monge - Ampère 方程:

$$\det(D^2 u) = f(x)$$

例 P-Laplace 方程: $\operatorname{div}(|\nabla u|^{p-2} \nabla u) = 0$ ($p > 2$)? 拟线性

$$\operatorname{div}(|\nabla u|^{p-2} \nabla u) = |\nabla u|^{p-2} \Delta u + \nabla(|\nabla u|^{p-2}) \cdot \nabla u$$

$$\begin{aligned} \text{其中 } \nabla(|\nabla u|^{p-2}) &= \nabla((\partial_1 u)^2 + (\partial_2 u)^2 + \dots + (\partial_n u)^2)^{\frac{p-2}{2}} \\ &= \frac{p-2}{2} (|\nabla u|^2)^{\frac{p-2}{2}-1} \begin{pmatrix} 2\partial_1 u \partial_{11} u + 2\partial_2 u \partial_{21} u + \dots + 2\partial_n u \partial_{n1} u \\ 2\partial_1 u \partial_{12} u + 2\partial_2 u \partial_{22} u + \dots + 2\partial_n u \partial_{n2} u \\ \vdots \\ 2\partial_1 u \partial_{1n} u + 2\partial_2 u \partial_{2n} u + \dots + 2\partial_n u \partial_{nn} u \end{pmatrix} \\ &= (p-2) |\nabla u|^{p-4} (\nabla u)^T D^2 u \end{aligned}$$

$$\text{于是 } \operatorname{div}(|\nabla u|^{p-2} \nabla u) = |\nabla u|^{p-2} \Delta u + (p-2) |\nabla u|^{p-4} (\nabla u)^T (D^2 u) (\nabla u)$$

§1.2 实例

1. Laplace 方程 $\Delta u = 0$

2. 特征值方程 $\Delta u + \lambda u = 0$ $\lambda \in \mathbb{R}$

3. 热方程 $u_t - a^2 \Delta u = 0$ $a > 0$

4. Schrödinger 方程 $u_t - i \Delta u = 0$

5. Kolmogorov 方程 $u_t - \sum_{i,j=1}^n a_{ij} u_{x_i x_j} + \sum_{i=1}^n b_i u_{x_i} = 0$

$$a_{ij}, b_i \in \mathbb{R} \quad u_{x_i x_j} = u_{j x_i} = \frac{\partial^2 u}{\partial x_i \partial x_j}$$

$$a_{ij}, b_i \in \mathbb{R} \quad u_{x_i x_j} = u_{ij} = \frac{\partial^2 u}{\partial x_i \partial x_j}$$

描述流体力学中非绝热流体运动规律的方程

6. Fokker-Planck eqn.

$$u_t - \sum_{i,j=1}^n (a_{ij} u)_{x_i x_j} + \sum_{i=1}^n (b_i u)_{x_i} = 0 \quad a_{ij}(x,t), b_i(x,t) \text{ 给定函数.}$$

$$a_{ij}, b_i \in \mathbb{R}$$

7. 输运方程: $u_t + \sum_{i=1}^n b_i u_{x_i} = 0 \quad b_i \in \mathbb{R}$

8. 波方程: $u_{tt} - a^2 \Delta u = 0 \quad a > 0$

非线性方程

Nonlinear poisson eqn. $\Delta u = u^2 - u$ 半线性 semi-linear.

极小曲面方程 $\operatorname{div} \left(\frac{\nabla u}{(1 + |\nabla u|^2)^{\frac{1}{2}}} \right) = 0$

Monge - Ampère eqn. $\det(D^2 u) = f(x)$

Hamilton - Jacobi eqn. $u_t + H(Du) = 0$ 完全非线性.

Burgers eqn. $u_t + u u_x = 0$ 拟线性

守恒律方程: $u_t + \operatorname{div} F(u) = 0$ 一阶拟线性 $\operatorname{div}(F(u)) = f'_1(u) \partial_1 u + f'_2(u) \partial_2 u + \dots + f'_n(u) \partial_n u$

多孔介质 eqn. $u_t - \Delta u^\gamma = 0 \quad \gamma > 1$

$$\Delta u^\gamma = \operatorname{div}(\nabla u^\gamma) = \operatorname{div}(\gamma u^{\gamma-1} \nabla u) = \gamma \nabla(u^{\gamma-1}) \cdot \nabla u + \gamma u^{\gamma-1} \Delta u$$

= 二阶拟线性.

KdV eqn. $u_t + u u_x + u_{xxx} = 0$ 水波 半线性

p-Laplace eqn. $\operatorname{div}(|\nabla u|^{p-2} \nabla u) = 0$ $p > 1, p \neq 2$ 拟线性, $p=2$ 线性.

Nonlinear wave eqn. $u_{tt} - a^2 \Delta u = f(u)$ 半线性

Boltzman eqn. $f_t + v \cdot D_x f = Q(f, f)$ 半线性

* 线性方程组: 设 $u(x) = (u_1(x), u_2(x), \dots, u_n(x))$

linear elastic sym. $\mu \Delta u + (\lambda + \mu) \nabla(\nabla \cdot u) = 0 \quad \mu, \lambda > 0$
(Lamé system)

$$u_t - \mu \Delta u - (\lambda + \mu) \nabla(\nabla \cdot u) = 0$$

Max well sym:
$$\begin{cases} \frac{1}{c} \frac{\partial E}{\partial t} = \nabla \times B \\ \frac{1}{c} \frac{\partial B}{\partial t} = -\nabla \times E \\ \nabla \cdot E = \nabla \cdot B = 0 \end{cases}$$

E electric field 电场 $E, B \in \mathbb{C}^3$
 B magnetic field 磁场.
 c 光速.

$$\nabla \times \left(\frac{1}{c} \frac{\partial B}{\partial t} \right) = -\nabla \times (\nabla \times E) = \Delta E + \nabla(\nabla \cdot E) = \Delta E$$

$$\parallel$$

$$\frac{1}{c} \frac{\partial}{\partial t} (\nabla \times B)$$

$$\parallel$$

$$\frac{1}{c^2} \frac{\partial}{\partial t} \left(\frac{\partial E}{\partial t} \right) = \frac{1}{c^2} \frac{\partial^2 E}{\partial t^2} \quad \frac{\partial^2 E}{\partial t^2} - c^2 \Delta E = 0$$

* 非线性方程组: 设 $u(x) = (u_1, u_2, \dots, u_n)$ 为向量值函数.

(1) 守恒律方程组 $u_t + (\vec{F}(u))_x = 0$ $(\vec{F}(u))_x = D_u \vec{F}(u) \cdot \vec{u}_x$ - 一阶拟线性

(2) 反应扩散方程组 $u_t - a^2 \Delta u = f(u)$ $a > 0$ 半线性

(3) Euler 方程组
$$\begin{cases} u_t + u \cdot \nabla u = -\nabla p \\ \nabla \cdot u = 0 \quad (\text{不可压}) \end{cases}$$
 - 一阶拟线性 $u \cdot \nabla u$ 对流项

u, p 表示流体速度和压力. (p 标量)

(4) Navier - Stokes sym.

$$\begin{cases} u_t + u \cdot \nabla u - \mu \Delta u = -\nabla p \\ \nabla \cdot u = 0 \quad (\text{不可压}) \end{cases}$$

= 阶半线性 $\mu \Delta u$ 粘性项 μ 粘性系数.

本书主要考虑三个典型的 PDE.

1. 位势方程 $-\Delta u = f(x)$ $u(x)$ $x \in \mathbb{R}^n$

2. 热方程 $u_t - a^2 \Delta u = f(x, t)$ $u(x, t)$ $(x, t) \in \mathbb{R}^n \times \mathbb{R}_+$

3. 波方程 $u_{tt} - a^2 \Delta u = f(x, t)$ $u(x, t)$

二阶线性 PDE 的一般形式:

$$\sum_{i,j=1}^m a_{ij}(x) u_{x_i x_j} + \sum_{i=1}^m b_i(x) u_{x_i} + c(x) u(x) = f(x) \quad (a_{ij} = a_{ji}) \quad (1.5)$$

其中 当 $m=n$ 时, $x = (x_1, \dots, x_n)$

当 $m=n+1$ 时, $x = (x_1, x_2, \dots, x_n, t)$

我们根据最高阶项的系数矩阵对二阶线性PDE进行分类:

$\left\{ \begin{array}{l} \text{椭圆} \\ \text{抛物} \\ \text{双曲} \end{array} \right.$

例如: 三个标准型

$$-\Delta u = f \quad \sim \quad A = (a_{ij})_{n \times n} = \begin{pmatrix} -1 & 0 & \dots & 0 \\ 0 & -1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & -1 \end{pmatrix} \quad b_i = 0 \quad c = 0$$

$$u_t - a^2 \Delta u = f \quad \sim \quad A = (a_{ij})_{(n+1) \times (n+1)} = \begin{pmatrix} -a^2 & 0 & \dots & 0 \\ 0 & -a^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix} \quad \begin{array}{l} b_i(x) = 0 \quad i=1, \dots, n \\ b_{n+1}(x) = 1 \end{array} \quad c = 0$$

$$u_t - a^2 \Delta u = f \quad \sim \quad A = (a_{ij})_{(n+1) \times (n+1)} = \begin{pmatrix} -a^2 & 0 & \dots & 0 \\ 0 & -a^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & -a^2 \\ 0 & 0 & \dots & 0 & 1 \end{pmatrix} \quad \begin{array}{l} b_i = 0 \quad (i=1, \dots, n+1), \quad c = 0 \end{array}$$

$$\text{对于 } \sum_{i,j=1}^m a_{ij}(x) u_{x_i x_j} + \sum_{i=1}^m b_i(x) u_{x_i} + c(x) u = f(x) \quad (1.5) \quad \begin{array}{l} \text{当 } m=n \text{ 时, } x=(x_1, \dots, x_n) \\ \text{当 } m=n+1 \text{ 时, } (x_{n+1}=t) \end{array}$$

$$\text{令系数矩阵 } A(x) := (a_{ij}(x)) \quad (a_{ij} = a_{ji})$$

设 $x_0 \in \mathbb{R}^n$ ($(x_0, t_0) \in \mathbb{R}^n \times \mathbb{R}_+$), 则 $A(x_0)$ 是实对称矩阵. (可对角化)

若 $A(x_0)$ 有 m 个负的特征值, 则称 (1.5) 在 x_0 处是椭圆型方程

若 $A(x_0)$ 有 $m-1$ 个负特征值, 1 个特征值为 0 则称 (1.5) 在 x_0 处是抛物型

若 $A(x_0)$ 有 $m-1$ 个负特征值, 1 个正特征值, 则 (1.5) 在 x_0 处是双曲型.

在区域 Ω 上是椭圆, 抛物, 双曲,

Thm 1.1 若 (1.5) 中二阶项系数矩阵 A 为常数矩阵, 且在 Ω 上是椭圆型(或双曲型)

则存在非奇异的变量替换, 可把 (1.5) 的二阶项变为标准形.

证. 以椭圆型方程为例

$$\sum_{i,j=1}^n a_{ij} u_{x_i x_j}(x) = f(x) - \sum_{i=1}^n b_i u_{x_i}(x) - c u(x) =: \tilde{f}(x, u(x), Du(x))$$

其中 $A := (a_{ij})_{n \times n}$ 实对称, $a_{ij} \in \mathbb{R}$, 则 $\exists$ 非奇异矩阵 B 以及 $|B|=1$ s.t.

$$BAB^T = -I$$

令 $y = Bx$, 其中 $B = (b_{ij})_{n \times n}$, 于是

$$\frac{\partial u}{\partial x_i} = \sum_{j=1}^n \frac{\partial u}{\partial y_j} \frac{\partial y_j}{\partial x_i} = \sum_{j=1}^n b_{ji} \frac{\partial u}{\partial y_j}$$

进而有

进而有

$$\nabla_x u = B^T \nabla_y u$$

$$\operatorname{div}_x = \operatorname{div}_y B$$

$$u(x) = u(B^{-1}y) =: \tilde{u}(y)$$

$$\sum_{i,j=1}^n a_{ij} u_{x_i x_j} = \operatorname{div}_x (A \nabla_x u) \stackrel{y=Bx}{=} \operatorname{div}_y (B A B^T \nabla_y \tilde{u}) = -\operatorname{div}_y (I \nabla_y \tilde{u}) = -\Delta_y \tilde{u}$$

则 (1.5) 可转化为 $-\Delta_y \tilde{u} = \tilde{f}(B^{-1}y, \tilde{u}(y), B^T D \tilde{u}(y))$.

$\mathbb{R}$

其中详细的计算过程:

关于坐标变换 $y=Bx$ 其中 $B=(b_{ij})_{n \times n}$, $|B|=1$

$$\text{由 } y=Bx = \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nn} \end{pmatrix} x$$

则

$$\frac{\partial u}{\partial x_i} = \sum_{j=1}^n \frac{\partial u}{\partial y_j} \frac{\partial y_j}{\partial x_i} = \sum_{j=1}^n b_{ji} \frac{\partial u}{\partial y_j}$$

具体地,

$$\frac{\partial u}{\partial x_1} = b_{11} \frac{\partial u}{\partial y_1} + b_{21} \frac{\partial u}{\partial y_2} + \cdots + b_{n1} \frac{\partial u}{\partial y_n}$$

$$\frac{\partial u}{\partial x_2} = b_{12} \frac{\partial u}{\partial y_1} + b_{22} \frac{\partial u}{\partial y_2} + \cdots + b_{n2} \frac{\partial u}{\partial y_n}$$

$\vdots$

$$\frac{\partial u}{\partial x_n} = b_{1n} \frac{\partial u}{\partial y_1} + b_{2n} \frac{\partial u}{\partial y_2} + \cdots + b_{nn} \frac{\partial u}{\partial y_n}$$

故

$$\nabla_x u(x) = B^T \nabla_y u(B^{-1}y)$$

以及

$$\nabla_x = B^T \nabla_y \quad \text{i.e.} \quad \begin{pmatrix} \partial_{x_1} \\ \partial_{x_2} \\ \vdots \\ \partial_{x_n} \end{pmatrix} = B^T \begin{pmatrix} \partial_{y_1} \\ \partial_{y_2} \\ \vdots \\ \partial_{y_n} \end{pmatrix}$$

取转置为

$$(\partial_{x_1}, \partial_{x_2}, \dots, \partial_{x_n}) = (\partial_{y_1}, \partial_{y_2}, \dots, \partial_{y_n}) B$$

设 $A=(a_{ij})_{n \times n}$ 为 n 阶项的系数矩阵

$$\begin{aligned} \operatorname{div}_x (A \nabla_x u(x)) &= (\partial_{x_1}, \partial_{x_2}, \dots, \partial_{x_n}) \begin{pmatrix} a_{11} \\ a_{12} \\ \vdots \\ a_{1n} \end{pmatrix}_{n \times n} \begin{pmatrix} \partial_{x_1} u(x) \\ \partial_{x_2} u(x) \\ \vdots \\ \partial_{x_n} u(x) \end{pmatrix} \\ &= (\partial_{y_1}, \partial_{y_2}, \dots, \partial_{y_n}) B A B^T \begin{pmatrix} \partial_{y_1} u(B^{-1}y) \\ \partial_{y_2} u(B^{-1}y) \end{pmatrix} \end{aligned}$$

$$= (\partial_{y_1}, \partial_{y_2}, \dots, \partial_{y_n}) B A B^T \begin{pmatrix} \partial_{y_1} u(B^T y) \\ \partial_{y_2} u(B^T y) \\ \vdots \\ \partial_{y_n} u(B^T y) \end{pmatrix}$$

$$= \operatorname{div}_y (B A B^T \nabla_y u(B^T y))$$